Reykjavik University

Machine Learning



Project 1 - Supervised Learning

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1 Introduction

Classifying colon tissue samples as tumour or normal based on gene expression has the potential to be a useful additional tool for diagnosing patients. Instead of relying only on manual examination, a model trained on previously collected data can recognise patterns in gene expression that indicate whether a tumour is there.

The data used in this project was originally collected by Alon et al. [1] and consists of 62 tissue samples, of which 40 are tumour tissue and 22 are normal tissue. Each sample is described by the expression levels of 2000 genes. We use the version of the dataset from the scikit-feature repository, in which the expression values have been discretised into three levels (stored as -2, 0 and 2), following the procedure described by Ding and Peng [3].

The dataset has been widely used to compare classification methods, and many studies report very high accuracies. For example, Guyon et al. [4] report 98% accuracy using only four genes selected by their method, compared to 86% for a baseline method. However, Ambroise and McLachlan [2] showed that such results are often too optimistic, because the genes used by the model were selected using all samples, including those later used to estimate the error. The goal of this project is therefore to find the best model for this dataset while obtaining an honest estimate of its performance, using nested cross-validation in which gene selection and hyperparameter tuning are done only on the training data.

2 Process

2.1 Data splitting

Nested cross validation was used because it protects against an overly optimistic performance estimate, which is easy to get with only 62 samples [2]. The outer loop (stratified 5-fold, repeated 3 times, giving 15 test folds) is used only to estimate performance, while an inner stratified 5-fold cross validation on each outer training set is used to choose the hyperparameters and the number of genes. The test samples are therefore never used for tuning or gene selection. A single hold out test set was not used, since 20% of the data is only about 12 samples. Stratification keeps the 40/22 class ratio in every fold, and the repeats reduce the dependence on how the samples are divided. A fixed random seed which is 42 makes the results reproducible.

2.2 Benchmarks

Two baselines were evaluated on the same outer folds as the other models. The first is a majority class classifier that always predicts "tumour". It reaches an accuracy of 64.6% but a balanced accuracy of only 0.5, and any useful model must clearly beat it. The second is a logistic regression with default settings, trained on all 2000 genes without gene selection or tuning. It shows how much the tuning and gene selection actually improve the results. As a reference from the literature, Ambroise and McLachlan [2] report a test error of around 17.5% (about 82.5% accuracy) for an SVM with gene selection when the selection bias is avoided.

2.3 Models

Six models were compared and chosen to cover both simple linear and more flexible non linear methods:

- **Logistic regression** (L1 or L2 penalty): a simple and interpretable linear baseline. The L1 penalty can also remove genes by itself.
- **Linear SVM**: has been successfully used for gene expression data [4], and its regularisation reduces overfitting when there are many more features than samples.
- **RBF SVM**: tests whether a non linear decision boundary helps.
- **k-nearest neighbours**: a distance based method but you can expected to suffer when many genes are irrelevant.
- **Naive Bayes** (categorical): suits the discrete gene levels, and performed well on this type of data in Ding and Peng [3].
- **Random forest**: a non linear ensemble method that is fairly robust to irrelevant features.

All models were implemented in scikit-learn [5].

2.4 Preprocessing

Every model is a pipeline that starts with gene selection (ANOVA F-test), keeping the k highest scoring genes. Because the selection is part of the pipeline, it is refitted on the training data in every fold, so the test samples never influence which genes are chosen [2]. Logistic regression, the SVMs and kNN are distance or margin based so the selected genes are standardised. For Naive Bayes the three levels are used directly as categories. To handle the class imbalance, "class_weight="balanced"" was included as a tuning option for logistic regression and the SVMs and always used for the random forest. Resampling methods such as SMOTE were not used since creating synthetic samples from only 22 normal samples in 2000 dimensions is unreliable.

2.5 Hyperparameters and runtime

Table 1 lists the values tried. For all models the number of genes was $k \in \{10, 50, 100, 500, \text{all}\}$. Regularisation parameters were searched on a logarithmic scale since their effect is multiplicative.

Model	Hyperparameters (besides k)	Configs
Logistic regression	$C \in \{0.01, \dots, 100\}$, penalty L1/L2, class weight	100
Linear SVM	$C \in \{10^{-4}, \dots, 1\}$, class weight	50
RBF SVM	$C \in \{0.1, \dots, 100\}$, $\gamma \in \{\text{scale}, 10^{-4}, 10^{-3}, 10^{-2}\}$, class weight	160
kNN	neighbours $\in \{1, 3, 5, 7, 9, 11, 15\}$, uniform/distance weights	70
Naive Bayes	smoothing $\alpha \in \{0.1, 0.5, 1, 2\}$	20
Random forest	max features $\in \{\sqrt{p}, 0.2\}$, max depth $\in \{\text{none}, 3\}$	20

Table 1: Hyperparameter grids searched in the inner cross validation.

Before running the experiment the runtime was estimated by timing one fit of each model on all genes and multiplying by the number of fits (configurations $\times$ 5 inner folds $\times$ 15 outer folds). This gave about 22 minutes on a single core, with the random forest (9.2 min) and Naive Bayes (5.6 min) being the most expensive. The random forest grid was therefore kept small. In practice the search ran in parallel and finished in a few minutes.

2.6 Evaluation metric

Balanced accuracy (the average of the recall for each class) was used to select and compare models. Plain accuracy is misleading here because always predicting “tumour” already gives 64.6% accuracy but only 0.5 balanced accuracy. Accuracy and ROC-AUC are also reported for completeness.

3 Results

Table 2 shows the nested cross-validation results for all models, averaged over the 15 outer test folds.

Model	Balanced accuracy	Accuracy	ROC-AUC
Logistic regression	**0.848** ± 0.085	0.835	0.877
Linear SVM	0.842 ± 0.105	0.855	0.884
RBF SVM	0.828 ± 0.107	0.840	0.875
Naive Bayes	0.818 ± 0.109	0.822	0.901
kNN	0.805 ± 0.128	0.812	0.870
Random forest	0.788 ± 0.087	0.808	0.910
Untuned logistic regression	0.833 ± 0.114	0.851	0.882
Majority class	0.500 ± 0.000	0.646	0.500

Table 2: Nested cross validation results (mean ± standard deviation over 15 outer folds).

All models clearly outperform the majority class baseline. Logistic regression achieved the highest balanced accuracy (0.848), closely followed by the linear SVM (0.842). However the differences between the four best models are around 0.03, which is much smaller than the standard deviations (about 0.09–0.11), so no model is clearly better than the others. The untuned logistic regression using all genes (0.833) performed almost as well as the tuned version. Figure 1 shows the spread over the outer folds. Some folds were difficult for all models. In outer fold 6, for example, every model scored between 0.48 and 0.74.

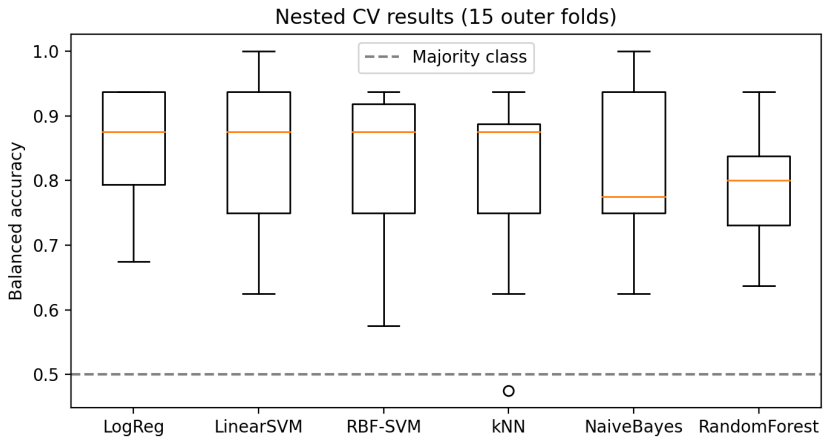


Figure 1: Balanced accuracy of each model over the 15 outer test folds.

Figure 2 shows how the number of selected genes affects each model. Logistic regression and the SVMs are stable across all values of k, while Naive Bayes and kNN drop sharply when all 2000 genes are used (Naive Bayes falls to about 0.61). Figure 3 shows that

logistic regression is only sensitive to very strong regularisation ($C = 0.01$); all values from $C = 0.1$ upwards give the same score. These two figures are based on a grid search on the full dataset and are only used to describe the sensitivity to hyperparameters, not to estimate performance.

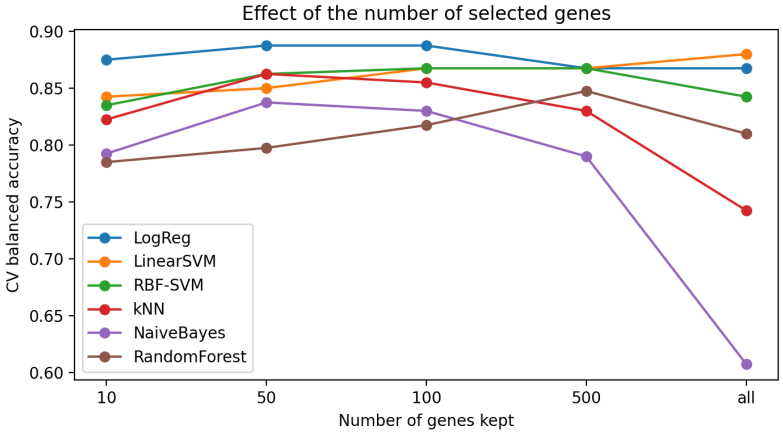


Figure 2: Cross validated balanced accuracy of the best configuration for each number of selected genes.

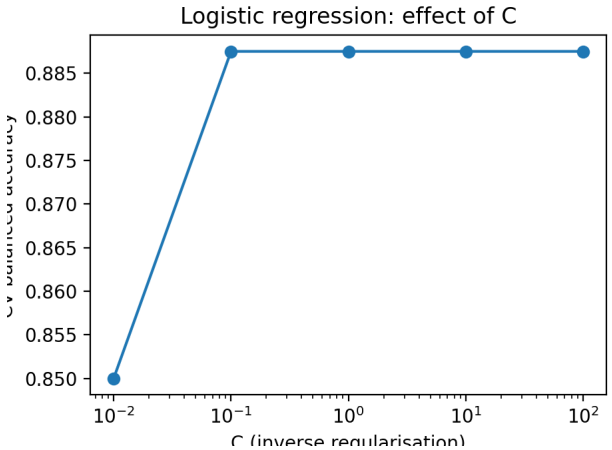


Figure 3: Effect of the regularisation parameter C on logistic regression.

The selected hyperparameters varied considerably between the outer folds. For logistic regression, for example, the number of selected genes ranged from 10 to all 2000, while $C = 0.1$ with an L2 penalty was the most common choice. Refitting the grid search on all 62 samples gave a final logistic regression model with $C = 100$, an L1 penalty, balanced class weights and all genes.

4 Conclusions

All six models were able to separate tumour from normal tissue far better than the majority class baseline, with the best models reaching a balanced accuracy of about 0.85. Logistic regression was chosen as the final model since it had the highest score and is the simpelst and most interpretable of the best models. The linear models performed at least as well as the non linear ones (RBF SVM, random forest), which suggests that the classes are close to linearly separable and that more complex models mostly add variance on such a small dataset.

The models differed in how sensitive they were to their hyperparameters. the regularised linear models were robust to both C and the number of genes, while Naive Bayes and kNN depended strongly on gene selection and performed poorly with all 2000 genes. The large variation in the selected hyperparameters between folds and the small difference between the tuned and untuned logistic regression, show that many settings perform about equally well and that tuning adds little on this dataset.

The results are lower than the 98% reported by Guyon et al. [4], but close to the test error of about 17.5% that Ambroise and McLachlan [2] found for the same method when the selection bias was removed. This supports the view that realistic accuracy on this dataset is around 80–85%. The remaining errors may partly be explained by the data itself: Ambroise and McLachlan [2] note that the labels of about six tissue samples are in doubt.

5 Future Work

With only 62 samples the results have high variance so more repeats of the outer cross validation or a larger external dataset would give a more reliable comparison between the models. Other gene selection methods could be tested, such as minimum redundancy selection [3] or recursive feature elimination [4], as well as the original continuous (non-discretised) expression values. Finally, removing or relabelling the samples with doubtful labels would show how much they affect the results.

References

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